The optimization problem is

$$\min f(x_1, x_2, x_3, x_4) = x_1^2 + \sin(x_3 - x_2) + x_4 + 1$$

subject to

$$\begin{aligned} x_1 + x_2 + x_3 &= 1, \\ x_1 - x_2 + x_4 &= 1. \end{aligned}$$

As there are two constraints, two variables have to be selected to be in the basis. The number of possibilities to select 2 variables out of 4 is 6. We consider each of them below.

1. One possible way to rewrite the constraints is to express x_3 and x_4 as a function of x_1 and x_2:

   $$\begin{aligned} x_3 &= 1 - x_1 - x_2, \\ x_4 &= 1 - x_1 + x_2. \end{aligned}$$

   The optimization problem can be rewritten so as to depend only on two variables x_1 and x_2:

   $$\min_{x \in \mathbb{R}^2} f(x_1, x_2) = x_1^2 + \sin(-x_1 - 2x_2 + 1) - x_1 + x_2 + 2,$$

   without constraint. Note the presence of the linear terms in x_1 and x_2. This function is unbounded.

   The basic variables are x_3 and x_4 and the non basic variables are x_1 and x_2.

2. A second option is to express x_1 and x_2 as a function of x_3 and x_4:

   $$\begin{aligned} x_1 &= 1 - \frac{x_3}{2} - \frac{x_4}{2}, \\ x_2 &= -\frac{x_3}{2} + \frac{x_4}{2}. \end{aligned}$$

   The optimization problem can be rewritten so as to depend only on two variables x_1 and x_2:

   $$\min f(x_3, x_4) = (1 - \frac{x_3}{2} - \frac{x_4}{2})^2 + \sin(\frac{3x_3}{2} - \frac{x_4}{2}) + x_4 + 1,$$

   without constraint. This function, again, is unbounded.

   The basic variables are x_1 and x_2 and the non basic variables are x_3 and x_4.

3. We can express x_1 and x_3 as a function of x_2 and x_4:

   $$\begin{aligned} x_1 &= 1 + x_2 - x_4, \\ x_3 &= -2x_2 + x_4. \end{aligned}$$

   $$\min f(x_2, x_4) = (1 + x_2 - x_4)^2 + \sin(-3x_2 + x_4) + x_4 + 1.$$

   We obtain again an unbounded function.

   The basic variables are x_1 and x_3, and the non basic variables are x_2 and x_4.

4. We can express x_1 and x_4 as a function of x_2 and x_3:

   $$\begin{aligned} x_1 &= 1 - x_2 - x_3, \\ x_4 &= 2x_2 + x_3. \end{aligned}$$

   We obtain

   $$\min f(x_2, x_3) = (1 - x_2 - x_3)^2 + \sin(x_3 - x_2) + 2x_2 + x_3 + 1,$$

   and the objective function is again unbounded.

   The basic variables are x_1 and x_4, and the non basic variables are x_2 and x_3.

5. We can express x_2 and x_3 as a function of x_1 and x_4:

$$x_2 = -1 + x_1 + x_4,$$
$$x_3 = 2 - 2x_1 - x_4.$$

We obtain

$$\min f(x_1, x_4) = x_1^2 + \sin(3 - 3x_1 - 2x_4) + x_4 + 1$$

and the objective function is again unbounded.

The basic variables are x_2 and x_3, and the non basic variables are x_1 and x_4.

6. We can express x_2 and x_4 as a function of x_1 and x_3:

$$x_2 = 1 - x_1 - x_3,$$
$$x_4 = 2 - 2x_1 - x_3.$$

We obtain

$$\min f(x_1, x_3) = x_1^2 + \sin(2x_3 - 1 + x_1) - 2x_1 - x_3 + 3,$$

and the objective function is again unbounded.

The basic variables are x_2 and x_4, and the non basic variables are x_1 and x_3.